100 Chemistry-Centred Problems in Mathematics for Machine Learning

Fully worked problems based on the mathematics-review materials

Coverage: vectors, matrices, tensors, systems of equations, rank, basis sets, eigenvalues, matrix decompositions, norms, inner products, projections, vector calculus, optimization, probability, statistics, and Gaussian models.

The problems follow the organization and terminology of the supplied mathematics-review slides and worked tutorials. Every problem places the mathematics in a chemical, materials, molecular-simulation, spectroscopy, thermodynamics, kinetics, or chemical-machine-learning setting.

Suggested use

- Problems 1–30: foundational linear algebra.
- Problems 31–50: analytic geometry and vector calculus.
- Problems 51–70: optimization and elementary probability.
- Problems 71–90: statistical inference and chemical machine learning.
- Problems 91–100: integrated and advanced applications.

Unless stated otherwise, all logarithms are natural logarithms and numerical answers are rounded reasonably.

Part I. Chemical Data as Vectors, Matrices, and Tensors

Problem 1 — Combining catalyst descriptor vectors

Two catalysts are represented by the descriptor vectors

$$\mathbf{x}_A = \begin{bmatrix} 1.2 \\ 2.0 \\ 0.5 \end{bmatrix}, \qquad \mathbf{x}_B = \begin{bmatrix} 0.8 \\ 1.5 \\ 0.7 \end{bmatrix},$$

where the entries are normalized electronegativity, coordination number, and adsorption-strength descriptors. A virtual catalyst is constructed as 25% catalyst A and 75% catalyst B. Find its descriptor vector.

Solution

A weighted combination is a vector linear combination:

$$\mathbf{x} = 0.25\mathbf{x}_A + 0.75\mathbf{x}_B.$$

Therefore,

$$\mathbf{x} = 0.25\begin{bmatrix}1.2\\2.0\\0.5\end{bmatrix} + 0.75\begin{bmatrix}0.8\\1.5\\0.7\end{bmatrix} = \begin{bmatrix}0.30+0.60\\0.50+1.125\\0.125+0.525\end{bmatrix} = \boxed{\begin{bmatrix}0.900\\1.625\\0.650\end{bmatrix}}.$$

The operation is legitimate because descriptor vectors live in a vector space closed under addition and scalar multiplication.

Problem 2 — Heat-capacity polynomials as vectors

The heat capacities of two species are represented by

$$C_{p,A}(T) = 20 + 0.020T + 1.0 \times 10^{-5}T^2,$$

$$C_{p,B}(T) = 15 + 0.010T + 2.0 \times 10^{-5}T^2.$$

Using the basis $\{1, T, T^2\}$, write both as coordinate vectors and find the polynomial for $2C_{p,A} - C_{p,B}$.

Solution

The coordinate vectors are

$$[C_{p,A}] = \begin{bmatrix}20\\0.020\\1.0\times10^{-5}\end{bmatrix}, \qquad [C_{p,B}] = \begin{bmatrix}15\\0.010\\2.0\times10^{-5}\end{bmatrix}.$$

Then

$$2[C_{p,A}] - [C_{p,B}] = \begin{bmatrix}40\\0.040\\2.0\times10^{-5}\end{bmatrix} - \begin{bmatrix}15\\0.010\\2.0\times10^{-5}\end{bmatrix} = \begin{bmatrix}25\\0.030\\0\end{bmatrix}.$$

Hence,

$$\boxed{2C_{p,A}(T) - C_{p,B}(T) = 25 + 0.030T.}$$

This illustrates that operations on polynomial functions become ordinary vector operations once a basis is chosen.

Problem 3 — Are normalized wavefunctions a vector space?

Let

$$S = \{\psi : \langle\psi|\psi\rangle = 1\}$$

be the set of normalized molecular wavefunctions. Determine whether S is a vector space.

Solution

A vector space must be closed under scalar multiplication. Take any $\psi \in S$. Then

$$\langle 2\psi|2\psi\rangle = 4\langle\psi|\psi\rangle = 4,$$

so $2\psi \notin S$. Therefore, S is not closed under scalar multiplication and is not a vector space.

It is also not closed under addition because $\psi + \psi = 2\psi$ is not normalized. The larger set of square-integrable, not-necessarily-normalized wavefunctions is a vector space. This is why an LCAO expansion

$$\phi = \sum_{\mu} c_{\mu}\chi_{\mu}$$

is a valid linear combination.

$$\boxed{S \text{ is not a vector space.}}$$

Problem 4 — Interpreting a chemical data matrix

A data set contains measurements for three molecules. Each molecule has four descriptors: molecular weight, dipole moment, HOMO energy, and polarizability. The data are arranged in a matrix $X \in \mathbb{R}^{3\times 4}$.

1. What does each row represent?
2. What does each column represent?
3. What is the dimension of the vector space containing X?

Solution

1. Each row is one molecule represented by four descriptors.
2. Each column is one descriptor evaluated for all three molecules.
3. A 3×4 matrix contains 12 scalar entries. The space $\mathbb{R}^{3\times 4}$ is therefore a 12-dimensional vector space.

$$\boxed{\dim\left(\mathbb{R}^{3\times 4}\right) = 12.}$$

The canonical basis consists of the 12 matrices E_{ij} having a 1 at position (i, j) and zeros elsewhere.

Problem 5 — Row-major and column-major storage

A transition-intensity matrix is

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Write the flattened sequence in:

1. row-major order;
2. column-major order.

Solution

Row-major storage reads each row from left to right:

$$\boxed{[1, 2, 3, 4, 5, 6]}.$$

Column-major storage reads each column from top to bottom:

$$\boxed{[1, 4, 2, 5, 3, 6]}.$$

In NumPy,

```
A.flatten()           # row-major: [1, 2, 3, 4, 5, 6]
A.flatten(order='F') # column-major: [1, 4, 2, 5, 3, 6]
```

Memory order matters when data are passed between Python and Fortran-based quantum-chemistry or linear-algebra libraries.

Problem 6 — Tensor ranks in molecular response theory

Classify the following objects by tensor rank and state the number of Cartesian components in three dimensions:

1. total energy E;
2. dipole moment μ_i;
3. polarizability α_{ij};
4. first hyperpolarizability β_{ijk}.

Solution

A tensor rank counts the number of indices required to identify one component.

1. E: rank 0, $3^0 = 1$ component.
2. μ_i: rank 1, $3^1 = 3$ components.
3. α_{ij}: rank 2, $3^2 = 9$ components before symmetry reduction.
4. β_{ijk}: rank 3, $3^3 = 27$ components before symmetry reduction.

$$\boxed{1,\ 3,\ 9,\ 27 \text{ components, respectively.}}$$

Tensor rank is distinct from matrix rank, which measures linear independence.

Problem 7 — Adding spectral matrices

Two independent absorbing species produce absorbance matrices

$$A_1 = \begin{bmatrix} 0.5 & 0.2 \\ 0.3 & 0.1 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 0.1 & 0.4 \\ 0.2 & 0.3 \end{bmatrix},$$

where rows are wavelengths and columns are samples. Assuming Beer–Lambert additivity, find the total absorbance matrix.

Solution

Matrix addition is elementwise:

$$A = A_1 + A_2 = \begin{bmatrix} 0.5 + 0.1 & 0.2 + 0.4 \\ 0.3 + 0.2 & 0.1 + 0.3 \end{bmatrix} = \boxed{\begin{bmatrix} 0.6 & 0.6 \\ 0.5 & 0.4 \end{bmatrix}}.$$

Both matrices must have the same shape because corresponding wavelengths and samples are being added.

Problem 8 — AO-to-MO matrix transformation

An AO Hamiltonian and an orthogonal coefficient matrix are

$$H_{\mathrm{AO}} = \begin{bmatrix} -1.0 & 0.2 \\ 0.2 & -0.5 \end{bmatrix}, \qquad C = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Compute

$$H_{\mathrm{MO}} = C^T H_{\mathrm{AO}} C.$$

Solution

Carrying out the two matrix multiplications gives

$$H_{\mathrm{MO}} = \boxed{\begin{bmatrix} -0.55 & -0.25 \\ -0.25 & -0.95 \end{bmatrix}}.$$

The corresponding index expression is

$$(H_{\mathrm{MO}})_{ij} = C_{\mu i} (H_{\mathrm{AO}})_{\mu\nu} C_{\nu j},$$

with repeated AO indices summed. Matrix multiplication must not be confused with the Hadamard product.

Problem 9 — Transposing a measurement matrix

A matrix contains two measured properties at three temperatures:

$$X = \begin{bmatrix} 1.0 & 10.0 \\ 1.5 & 12.0 \\ 2.0 & 15.0 \end{bmatrix}.$$

Rows are temperatures and columns are viscosity and conductivity. Find X^T and interpret its rows.

Solution

$$X^T = \boxed{\begin{bmatrix} 1.0 & 1.5 & 2.0 \\ 10.0 & 12.0 & 15.0 \end{bmatrix}}.$$

After transposition, the first row contains viscosity at all temperatures and the second row contains conductivity at all temperatures. Transposition exchanges observation and feature axes.

Problem 10 — Inverting a two-sensor calibration

Two sensors respond to concentrations c_1 and c_2 according to

$$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 & 0.2 \\ 0.1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

If $s_1 = 1.4$ and $s_2 = 2.1$, determine the concentrations.

Solution

Let

$$A = \begin{bmatrix} 1 & 0.2 \\ 0.1 & 1 \end{bmatrix}.$$

Its determinant is

$$\det A = 1 - 0.02 = 0.98 \neq 0,$$

so the calibration is invertible. Solving $A\mathbf{c} = \mathbf{s}$ gives

$$\boxed{c_1 = 1.0, \qquad c_2 = 2.0.}$$

Verification:

$$1.0 + 0.2(2.0) = 1.4, \qquad 0.1(1.0) + 2.0 = 2.1.$$

Part II. Linear Equations, Stoichiometry, Rank, and Least Squares

Problem 11 — Balancing propane combustion as a null-space problem

Balance $C_3H_8 + O_2 \to CO_2 + H_2O$ by treating the stoichiometric coefficients as a null-space vector.

Solution

Let the coefficients of C_3H_8, O_2, CO_2, H_2O be x_1, x_2, x_3, x_4. Conservation of C, H, and O gives

$$3x_1 - x_3 = 0, \qquad 8x_1 - 2x_4 = 0, \qquad 2x_2 - 2x_3 - x_4 = 0.$$

Thus $x_3 = 3x_1$, $x_4 = 4x_1$, and $x_2 = 5x_1$. Choosing $x_1 = 1$ gives

$$\boxed{C_3H_8 + 5O_2 \to 3CO_2 + 4H_2O.}$$

The null space is one-dimensional, so the equation is unique up to overall scaling.

Problem 12 — Chemical production planning

A plant has 4 units of feed A, 3 units of feed B, and 3 units of feed C. Product amounts satisfy

$$x_1 + x_2 + x_3 = 4, \qquad x_1 + x_2 = 3, \qquad x_2 + 2x_3 = 3.$$

Find x_1, x_2, x_3.

Solution

Subtracting the second equation from the first gives $x_3 = 1$. The third equation then gives $x_2 = 1$, and the second gives $x_1 = 2$. Therefore,

$$\boxed{(x_1, x_2, x_3) = (2, 1, 1).}$$

Problem 13 — Two-component Beer–Lambert deconvolution

Solve

$$\begin{bmatrix} 1.0 & 0.5 \\ 0.2 & 1.0 \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}.$$

Solution

The equations are $c_A + 0.5c_B = 1.2$ and $0.2c_A + c_B = 0.8$. Substitution gives $0.9c_B = 0.56$, so $c_B = 0.6222$. Then $c_A = 1.2 - 0.5c_B = 0.8889$. Hence,

$$\boxed{c_A \approx 0.889, \qquad c_B \approx 0.622.}$$

Problem 14 — Steady-state concentration in a consecutive reaction

For $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, take $[A] = 2.0$, $k_1 = 0.30\ \mathrm{s}^{-1}$, and $k_2 = 0.20\ \mathrm{s}^{-1}$. Under the steady-state approximation for B, find $[B]$.

Solution

At steady state,

$$0 = \frac{d[B]}{dt} = k_1[A] - k_2[B].$$

Therefore,

$$[B] = \frac{k_1[A]}{k_2} = \frac{0.30(2.0)}{0.20} = \boxed{3.0\ \mathrm{mol\,L^{-1}}}.$$

Problem 15 — A reaction-extent system with a free variable

Solve

$$x_1 + 2x_2 + 3x_3 = 3, \qquad 2x_1 + 4x_2 + 3x_3 = 8.$$

Solution

Subtracting twice the first equation from the second gives $-3x_3 = 2$, so $x_3 = -2/3$. The first equation becomes $x_1 + 2x_2 = 5$. Let $x_2 = t$. Then

$$\boxed{\mathbf{x} = \begin{bmatrix} 5 \\ 0 \\ -2/3 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}.}$$

The second vector spans the null space.

Problem 16 — Consistency controlled by a parameter

For what value of a is the system

$$x + y = 1, \qquad 2x + 2y = a$$

consistent? How many solutions exist then?

Solution

Twice the first equation gives $2x + 2y = 2$, so consistency requires $\boxed{a = 2}$. At that value the second equation is redundant, leaving infinitely many solutions:

$$\boxed{(x, y) = (1, 0) + t(-1, 1).}$$

For $a \neq 2$, no solution exists.

Problem 17 — Inverting a calorimeter calibration matrix

Given

$$\begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix},$$

find the inverse calibration and determine m_1, m_2 for $(q_1, q_2) = (5, 3)$.

Solution

The determinant is 1, and

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = A^{-1} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \boxed{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}.$$

Problem 18 — Arrhenius parameters from two measurements

Suppose $\ln k = 0$ at 300 K and $\ln k = 2$ at 600 K. Using $\ln k = \ln A - E_a/(RT)$, determine E_a and A.

Solution

The slope of $\ln k$ against $1/T$ is

$$m = \frac{2}{1/600 - 1/300} = -1200 \text{ K}.$$

Thus $E_a = -mR = 1200(8.314) = 9.98 \text{ kJ mol}^{-1}$. At 300 K, $0 = \ln A - 4$, so $\ln A = 4$ and $A = e^4 \approx 54.6$.

$$\boxed{E_a \approx 9.98 \text{ kJ mol}^{-1}, \qquad A \approx 54.6.}$$

Problem 19 — Pseudoinverse concentration fitting

Find the least-squares solution of

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} \approx \begin{bmatrix} 1.1 \\ 2.0 \\ 0.9 \end{bmatrix}.$$

Solution

The normal equations are

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = \begin{bmatrix} 3.1 \\ 2.9 \end{bmatrix}.$$

Solving gives

$$\boxed{c_A = 1.1, \qquad c_B = 0.9.}$$

Problem 20 — Number of independent reactions

The reaction vectors are

$$\mathbf{r}_1 = (-1, 1, 0)^T, \quad \mathbf{r}_2 = (0, -1, 1)^T, \quad \mathbf{r}_3 = (-1, 0, 1)^T.$$

How many are independent?

Solution

Because $\mathbf{r}_3 = \mathbf{r}_1 + \mathbf{r}_2$, the third reaction is dependent. The reaction matrix has rank 2.

$$\boxed{\text{Two reactions are linearly independent.}}$$

Part III. Basis Sets, Linear Mappings, Eigenvalues, and Matrix Decompositions

Problem 21 — Hybrid-orbital vectors as a basis

Do $\mathbf{h}_1 = (1,1,0)^T$, $\mathbf{h}_2 = (1,-1,0)^T$, and $\mathbf{h}_3 = (0,0,1)^T$ form a basis of a three-dimensional AO coefficient space?

Solution

Using the vectors as columns gives

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det H = -2 \neq 0.$$

The vectors are independent, and three independent vectors in a three-dimensional space form a basis.

$$\boxed{\{\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3\} \text{ is a basis.}}$$

Problem 22 — Coordinates in a new descriptor basis

Express $\mathbf{x} = (2,3)^T$ in the basis $\mathbf{b}_1 = (1,0)^T$, $\mathbf{b}_2 = (1,1)^T$.

Solution

Solve $\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 = \mathbf{x}$. The second component gives $\alpha_2 = 3$, and the first gives $\alpha_1 = -1$.

$$\boxed{[\mathbf{x}]_{\mathcal{B}} = (-1,3)^T.}$$

Only the coordinates change; the underlying chemical point does not.

Problem 23 — Differentiation as a matrix

In the basis $\{1, x, x^2\}$, construct the derivative matrix and apply it to $f(x) = 1 - 2x + 3x^2$.

Solution

The images of the basis functions are 0, 1, and $2x$, so

$$D = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

With $[f] = (1, -2, 3)^T$,

$$D[f] = (-2, 6, 0)^T,$$

which represents

$$\boxed{f'(x) = -2 + 6x.}$$

Problem 24 — Image and kernel of a descriptor projection

For

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

find $\mathrm{Im}(P)$ and $\ker(P)$.

Solution

$P(x_1, x_2, x_3)^T = (x_1, x_2, 0)^T$. Therefore,

$$\boxed{\mathrm{Im}(P) = \mathrm{span}\{(1, 0, 0)^T, (0, 1, 0)^T\}},$$

$$\boxed{\ker(P) = \mathrm{span}\{(0, 0, 1)^T\}.}$$

Also $P^2 = P$, confirming that it is a projection.

Problem 25 — Eigenmodes of two-state exchange

For

$$K = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix},$$

find the eigenvalues and interpret the modes.

Solution

$\det(K - \lambda I) = \lambda(\lambda + 2)$, so the eigenvalues are 0 and -2. The eigenvector for 0 is $(1,1)^T$, representing conserved total population. The eigenvector for -2 is $(1,-1)^T$, representing a population difference that decays as e^{-2t}.

$$\boxed{\lambda = 0 \text{ and } -2.}$$

Problem 26 — Hückel orbitals of ethylene

Diagonalize

$$H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Solution

The eigenvalues are $+1$ and -1. Corresponding normalized eigenvectors are

$$\boxed{\mathbf{v}_+ = \frac{1}{\sqrt{2}}(1,1)^T, \qquad \mathbf{v}_- = \frac{1}{\sqrt{2}}(1,-1)^T.}$$

They represent bonding and antibonding combinations, respectively.

Problem 27 — Normal modes from a Hessian

For the mass-weighted Hessian

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

find the normal modes.

Solution

The eigenvalues are 3 and 1, with normalized eigenvectors $(1,1)^T/\sqrt{2}$ and $(1,-1)^T/\sqrt{2}$. The frequency ratio is

$$\omega_1 : \omega_2 = \sqrt{3} : 1.$$

The first mode is in phase and the second is out of phase.

Problem 28 — Cholesky factorization of a covariance matrix

Factor

$$\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} = LL^T$$

with lower-triangular L.

Solution

Let $L = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$. Matching entries gives $a = 2$, $b = 1$, and $c = \sqrt{2}$.

$$\boxed{L = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix}.}$$

Problem 29 — Singular values and retained signal

For

$$X = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

find the singular values and the fraction of squared signal retained by the largest one.

Solution

The singular values are 3 and 1. The retained fraction is

$$\frac{3^2}{3^2 + 1^2} = \frac{9}{10} = 0.90.$$

$$\boxed{\text{90\% of the squared spectral signal is retained.}}$$

Problem 30 — Best rank-1 spectral approximation

Find the best rank-1 approximation to

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Solution

The dominant eigenvalue is 4 with normalized eigenvector $(1,1)^T/\sqrt{2}$. Therefore,

$$X_1 = 4\mathbf{u}_1\mathbf{u}_1^T = \boxed{\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}}.$$

The discarded singular value is 2, so the Frobenius error is 2.

Part IV. Norms, Inner Products, Orthogonality, and Projections

Problem 31 — Comparing L_1 and L_2 descriptor norms

For $\mathbf{x} = (3, -4, 0)^T$, calculate the Manhattan and Euclidean norms.

Solution

$$\|\mathbf{x}\|_1 = |3| + |-4| + 0 = 7,$$

$$\|\mathbf{x}\|_2 = \sqrt{3^2 + (-4)^2} = 5.$$

$$\boxed{\|\mathbf{x}\|_1 = 7, \qquad \|\mathbf{x}\|_2 = 5.}$$

The two norms encode different geometries for chemical similarity.

Problem 32 — Nearest molecule depends on the metric

A query molecule is $Q = (0, 0)$. Library molecules are $A = (3, 0)$ and $B = (2, 2)$. Determine the nearest under L_1 and L_2 distance.

Solution

For A, $d_1 = 3$ and $d_2 = 3$. For B, $d_1 = 4$ and $d_2 = \sqrt{8} \approx 2.83$.

$$\boxed{A \text{ is nearest under } L_1; \quad B \text{ is nearest under } L_2.}$$

Problem 33 — Weighted inner product for scaled descriptors

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{x} = (1, 1)^T, \quad \mathbf{y} = (2, -1)^T.$$

Using $\langle x, y \rangle_A = x^T A y$, find the inner product and induced norms.

Solution

$Ay = (2, -4)^T$, so

$$\langle x, y \rangle_A = -2.$$

Also,

$$\|x\|_A = \sqrt{x^T A x} = \sqrt{5}, \qquad \|y\|_A = \sqrt{y^T A y} = \sqrt{8}.$$

$$\boxed{\langle x, y \rangle_A = -2, \quad \|x\|_A = \sqrt{5}, \quad \|y\|_A = \sqrt{8}.}$$

Problem 34 — Angle between reaction fingerprints

For $\mathbf{r}_1 = (1, 1, 0)^T$ and $\mathbf{r}_2 = (1, 0, 1)^T$, find the angle under the dot product.

Solution

$$\cos\theta = \frac{\mathbf{r}_1^T\mathbf{r}_2}{\|\mathbf{r}_1\|\|\mathbf{r}_2\|} = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

Therefore,

$$\boxed{\theta = 60^\circ.}$$

Problem 35 — Orthogonality of orbital-like functions

Using $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$, determine whether $f(x) = 1$ and $g(x) = x$ are orthogonal.

Solution

$$\langle 1, x\rangle = \int_{-1}^{1} x\,dx = 0.$$

Hence,

$$\boxed{1 \text{ and } x \text{ are orthogonal on } [-1, 1].}$$

Problem 36 — Gram–Schmidt orthogonalization

Orthogonalize $\mathbf{b}_1 = (1, 1)^T$ and $\mathbf{b}_2 = (1, 0)^T$.

Solution

Normalize the first vector:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)^T.$$

Remove the projection of b_2 along u_1:

$$\mathbf{v}_2 = \mathbf{b}_2 - (\mathbf{u}_1^T\mathbf{b}_2)\mathbf{u}_1 = (1/2, -1/2)^T.$$

Normalize:

$$\boxed{\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)^T.}$$

Problem 37 — Orthogonal complement of a conservation plane

The allowed composition changes satisfy $x + y + z = 0$. Find the orthogonal complement.

Solution

The vector $(1, 1, 1)^T$ is normal to the plane and orthogonal to every allowed change. Therefore,

$$\boxed{U^{\perp} = \operatorname{span}\{(1, 1, 1)^T\}.}$$

Problem 38 — Projection onto a stoichiometric direction

Project $\mathbf{x} = (2, 0, 1)^T$ onto the line spanned by $\mathbf{b} = (1, -1, 0)^T$.

Solution

$$\operatorname{proj}_b x = \frac{b^T x}{b^T b} b = \frac{2}{2} b = (1, -1, 0)^T.$$

The residual is

$$x - \operatorname{proj}_b x = (1, 1, 1)^T.$$

Thus,

$$\boxed{x_{\parallel} = (1, -1, 0)^T, \quad x_{\perp} = (1, 1, 1)^T.}$$

Problem 39 — Projection matrix onto a chemical line

Find the projection matrix onto $\operatorname{span}(1, 1)^T$ and project $x = (3, 1)^T$.

Solution

$$P = b(b^T b)^{-1} b^T = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Then

$$Px = \frac{1}{2}(4, 4)^T = \boxed{(2, 2)^T}.$$

Problem 40 — Least-squares calibration through the origin

A detector is modeled by $y = mx$ using data $(1, 2.0)$, $(2, 4.1)$, and $(3, 5.9)$. Find m.

Solution

$$m = \frac{x^T y}{x^T x} = \frac{1(2.0) + 2(4.1) + 3(5.9)}{1^2 + 2^2 + 3^2} = \frac{27.9}{14}.$$

$$\boxed{m \approx 1.993.}$$

Part V. Gradients, Jacobians, Hessians, and Taylor Expansions

Problem 41 — Gradient of a potential-energy surface

For

$$E(x, y) = (x - 1)^2 + 2(y + 1)^2,$$

find ∇E and the direction of steepest energy decrease at $(0, 0)$.

Solution

$$\nabla E = (2(x - 1), 4(y + 1))^T.$$

At the origin, $\nabla E = (-2, 4)^T$. The direction of steepest decrease is the negative gradient:

$$\boxed{-\nabla E(0, 0) = (2, -4)^T.}$$

Problem 42 — Gradient of a force-model loss

For

$$L(w) = \frac{1}{2}[(w - 2)^2 + (2w - 5)^2],$$

find dL/dw and evaluate it at $w = 2$.

Solution

$$\frac{dL}{dw} = (w-2) + 2(2w-5) = 5w - 12.$$

At $w = 2$,

$$\boxed{L'(2) = -2.}$$

The negative gradient indicates that increasing w will initially decrease the loss.

Problem 43 — Chain rule along a reaction coordinate

Let $x_1 = t$, $x_2 = t^2$, and $E = x_1^2 + 3x_2$. Find dE/dt.

Solution

$$\nabla E = (2x_1, 3)^T, \qquad \frac{d\mathbf{x}}{dt} = (1, 2t)^T.$$

Therefore,

$$\frac{dE}{dt} = \nabla E^T \frac{d\mathbf{x}}{dt} = 2t + 6t = \boxed{8t}.$$

Problem 44 — Jacobian from Cartesian to bond coordinates

Define $r = x_2 - x_1$ and $c = (x_1 + x_2)/2$. Find $\partial(r, c)/\partial(x_1, x_2)$.

Solution

$$\boxed{J = \begin{bmatrix} -1 & 1 \\ 1/2 & 1/2 \end{bmatrix}.}$$

The first row describes stretching, and the second describes translation of the molecular centre.

Problem 45 — Jacobian of nonlinear kinetic observables

Let $f_1(A, k) = kA$ and $f_2(A, k) = A/(1 + A)$. Find the Jacobian at $A = 1$, $k = 2$.

Solution

$$J = \begin{bmatrix} k & A \\ (1+A)^{-2} & 0 \end{bmatrix}.$$

Thus,

$$\boxed{J(1,2) = \begin{bmatrix} 2 & 1 \\ 1/4 & 0 \end{bmatrix}.}$$

Problem 46 — Derivative of a response matrix

For

$$A(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_1^2 \end{bmatrix},$$

find the two partial derivative matrices.

Solution

$$\boxed{\frac{\partial A}{\partial x_1} = \begin{bmatrix} 1 & 0 \\ 0 & 2x_1 \end{bmatrix}, \qquad \frac{\partial A}{\partial x_2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.}$$

Together they form a rank-3 derivative tensor.

Problem 47 — Hessian classification of a stationary point

For $f(x, y) = x^2 - y^2$, find the Hessian and classify $(0, 0)$.

Solution

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Its eigenvalues have opposite signs, so the Hessian is indefinite.

$$\boxed{(0,0) \text{ is a saddle point.}}$$

This is analogous to a transition state with one unstable direction.

Problem 48 — Frequencies from a mass-weighted Hessian

For $H = \mathrm{diag}(4, 9)$, find the harmonic angular frequencies.

Solution

The Hessian eigenvalues equal ω_i^2. Therefore,

$$\boxed{\omega_1 = 2, \qquad \omega_2 = 3}$$

in the chosen units.

Problem 49 — Taylor approximation of an Arrhenius-like function

Let $k(T) = \exp(-1000/T)$. Use a first-order Taylor expansion around 500 K to estimate $k(510\text{ K})$.

Solution

$k(500) = e^{-2} = 0.135335$ and

$$k'(500) = e^{-2}\frac{1000}{500^2} = 0.00054134.$$

Therefore,

$$k(510) \approx 0.135335 + 10(0.00054134) = \boxed{0.14075}.$$

The exact value is about 0.14076.

Problem 50 — Linearization of protonation fraction

For

$$q(\mathrm{pH}) = \frac{1}{1 + 10^{\mathrm{pH}-\mathrm{p}K_a}},$$

estimate q at $\mathrm{pH} = \mathrm{p}K_a + 0.1$ by linearization around $\mathrm{p}K_a$.

Solution

At pH $=$ pK_a, $q = 1/2$. Also,

$$q' = -\ln(10)q(1-q) = -\frac{\ln 10}{4} \approx -0.57565.$$

Hence,

$$q \approx 0.5 - 0.57565(0.1) = \boxed{0.4424}.$$

The exact value is approximately 0.443.

Part VI. Optimization in Chemical Models

Problem 51 — Gradient descent on a one-dimensional energy

For $E(x) = (x-3)^2$, start from $x_0 = 0$ and use step size $\gamma = 0.25$. Perform three gradient-descent updates.

Solution

$E'(x) = 2(x-3)$ and $x_{n+1} = x_n - 0.25E'(x_n)$. Therefore,

$$x_1 = 1.5, \qquad x_2 = 2.25, \qquad x_3 = 2.625.$$

$$\boxed{x_1 = 1.5,\ x_2 = 2.25,\ x_3 = 2.625.}$$

The sequence converges to the minimum at $x = 3$.

Problem 52 — Why a large step size diverges

Use the same energy $E(x) = (x-3)^2$, but take $\gamma = 1.1$. Explain the behavior.

Solution

The error evolves as

$$x_{n+1} - 3 = (1-2\gamma)(x_n - 3).$$

Here $1 - 2\gamma = -1.2$, whose magnitude exceeds 1. The error alternates sign and grows by 20% each step.

$$\boxed{\text{The iteration diverges because } |1 - 2\gamma| = 1.2 > 1.}$$

Problem 53 — Momentum on a quadratic energy

Minimize $f(x) = x^2/2$ using

$$v_{n+1} = 0.5v_n - 0.2f'(x_n), \qquad x_{n+1} = x_n + v_{n+1},$$

with $x_0 = 4$, $v_0 = 0$. Compute two steps.

Solution

Since $f'(x) = x$,

$$v_1 = -0.8, \quad x_1 = 3.2,$$

$$v_2 = 0.5(-0.8) - 0.2(3.2) = -1.04, \quad x_2 = 2.16.$$

$$\boxed{(x_1, v_1) = (3.2, -0.8), \quad (x_2, v_2) = (2.16, -1.04).}$$

Problem 54 — Two stochastic-gradient updates

For $L_i(w) = \frac{1}{2}(wx_i - y_i)^2$, use samples $(1, 2)$ and $(2, 4)$ sequentially. Start at $w_0 = 0$ with $\gamma = 0.1$.

Solution

The sample gradient is $(wx_i - y_i)x_i$. For the first sample, $g_1 = -2$, so $w_1 = 0.2$. For the second, $g_2 = (0.4 - 4)2 = -7.2$, so $w_2 = 0.92$.

$$\boxed{w_1 = 0.20, \qquad w_2 = 0.92.}$$

Problem 55 — Minimum energy under a composition constraint

Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$.

Solution

The Lagrangian is $\mathcal{L} = x^2 + y^2 + \lambda(x + y - 1)$. Stationarity gives $2x + \lambda = 0$ and $2y + \lambda = 0$, hence $x = y$. The constraint gives

$$\boxed{x = y = \frac{1}{2}.}$$

The minimum value is 1/2.

Problem 56 — Active nonnegativity constraint

Minimize

$$f(x, y) = (x - 1)^2 + (y + 1)^2$$

subject to $x + y = 1$, $x \geq 0$, and $y \geq 0$.

Solution

Ignoring nonnegativity, substitution $y = 1 - x$ gives the optimum $(x, y) = (1.5, -0.5)$, which is infeasible. Therefore, the boundary $y = 0$ is active, and the equality constraint gives $x = 1$.

$$\boxed{x = 1, \qquad y = 0.}$$

Problem 57 — Optimal binary catalyst composition

Maximize

$$S(x, y) = 4x + 3y - x^2 - y^2$$

subject to $x + y = 2$.

Solution

Set $y = 2 - x$. Then

$$S(x) = 2 + 5x - 2x^2,$$

so $S'(x) = 5 - 4x = 0$. Thus $x = 1.25$ and $y = 0.75$. Since $S'' = -4 < 0$, this is a maximum.

$$\boxed{x = 1.25, \qquad y = 0.75.}$$

Problem 58 — Least-squares kinetic fit with intercept

Fit $y = mx + c$ to $(1, 2)$, $(2, 3)$, and $(3, 5)$.

Solution

With

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix},$$

the normal equations give

$$\boxed{m = 1.5, \qquad c = \frac{1}{3}.}$$

Thus $\hat{y} = 1.5x + 0.333$.

Problem 59 — Closest geometry on a bond-length constraint

Find the point on $x^2 + y^2 = 1$ closest to $(2, 0)$.

Solution

The nearest point lies on the radial line from the origin to $(2, 0)$ and on the unit circle. Therefore,

$$\boxed{(x, y) = (1, 0).}$$

A Lagrange-multiplier calculation gives the same result.

Problem 60 — Minimum-cost feed blend

Feed X costs 2 units per mole and feed Y costs 3 units per mole. At least 10 total moles are required, with $x \le 6$, $y \le 8$, and $x, y \ge 0$. Minimize $C = 2x + 3y$.

Solution

Use as much of the cheaper feed X as possible: $x = 6$. The remaining requirement is $y = 4$. Therefore,

$$\boxed{x = 6, \qquad y = 4, \qquad C_{\text{min}} = 24.}$$

The optimum lies at a corner of the feasible region.

Part VII. Probability Spaces and Chemical Random Variables

Problem 61 — Isotope count as a random variable

A molecule contains three equivalent atoms. Each is independently a heavy isotope with probability 0.2. Let X be the number of heavy isotopes. State the sample space and target space.

Solution

Using L and H,

$$\Omega = \{LLL, LLH, LHL, HLL, LHH, HLH, HHL, HHH\}.$$

The random variable counts H symbols, so

$$\boxed{X(\Omega) = \{0, 1, 2, 3\}.}$$

The sample space contains isotope arrangements; the target space contains counts.

Problem 62 — Union and complement of reactor alarms

Let $P(A) = 0.10$, $P(B) = 0.20$, and $P(A \cap B) = 0.05$ for high-temperature and high-pressure alarms. Find the probability of at least one alarm and of neither alarm.

Solution

$$P(A \cup B) = 0.10 + 0.20 - 0.05 = 0.25.$$

Therefore,

$$P(\text{neither}) = 1 - 0.25 = 0.75.$$

$$\boxed{P(\text{at least one}) = 0.25, \qquad P(\text{neither}) = 0.75.}$$

Problem 63 — Conditional probability of reaction success

Suppose $P(S \cap H) = 0.18$ and $P(H) = 0.30$, where H denotes high catalyst loading and S denotes successful conversion. Find $P(S|H)$.

Solution

$$P(S|H) = \frac{P(S \cap H)}{P(H)} = \frac{0.18}{0.30} = \boxed{0.60}.$$

Problem 64 — Independent versus mutually exclusive events

Two deactivation mechanisms have probabilities $P(A) = 0.10$ and $P(B) = 0.20$. Assuming independence, find $P(A \cap B)$ and state whether the mechanisms are mutually exclusive.

Solution

$$P(A \cap B) = P(A)P(B) = 0.02.$$

Because the intersection probability is nonzero, they can occur together.

$$\boxed{P(A \cap B) = 0.02; \quad A \text{ and } B \text{ are not mutually exclusive.}}$$

Problem 65 — Number of active catalytic sites

A nanoparticle has five independent sites, each active with probability 0.8. Find the probability that exactly four are active.

Solution

$$P(X = 4) = \binom{5}{4}(0.8)^4(0.2) = 5(0.4096)(0.2).$$

$$\boxed{P(X = 4) = 0.4096.}$$

Problem 66 — Polymer bond scission

A chain has six monomers and five bonds. Each bond breaks independently with probability 0.1. Find the probability of exactly two chain segments.

Solution

Two segments require exactly one broken bond:

$$P = \binom{5}{1}(0.1)(0.9)^4 = 5(0.1)(0.6561).$$

$$\boxed{P \approx 0.3281.}$$

Problem 67 — PMF and CDF for product count

Two independent active sites each form product with probability 0.3. Let X be the number of products. Find the PMF and $P(X \leq 1)$.

Solution

$$P(X = 0) = 0.7^2 = 0.49,$$

$$P(X = 1) = 2(0.3)(0.7) = 0.42,$$

$$P(X = 2) = 0.3^2 = 0.09.$$

Thus,

$$\boxed{P(X \leq 1) = 0.49 + 0.42 = 0.91.}$$

Problem 68 — Continuous residence-time probability

A residence time follows an exponential distribution with rate $\lambda = 0.5\ \text{min}^{-1}$. Find $P(T < 2\ \text{min})$.

Solution

For an exponential variable,

$$F(t) = 1 - e^{-\lambda t}.$$

Therefore,

$$P(T < 2) = 1 - e^{-1} = \boxed{0.6321 \text{ approximately.}}$$

Problem 69 — Marginal and conditional conformer probability

A joint distribution is

	Reactive	Nonreactive
Conformer A	0.30	0.20
Conformer B	0.10	0.40

Find $P(R)$ and $P(A|R)$.

Solution

$$P(R) = 0.30 + 0.10 = 0.40.$$

$$P(A|R) = \frac{0.30}{0.40} = 0.75.$$

$$\boxed{P(R) = 0.40, \qquad P(A|R) = 0.75.}$$

Problem 70 — Bayesian diagnosis of catalyst failure

A catalyst has prior failure probability 0.05. A diagnostic is positive with probability 0.90 for failed catalysts and 0.10 for healthy catalysts. Find $P(F|+)$.

Solution

$$P(F|+) = \frac{0.90(0.05)}{0.90(0.05) + 0.10(0.95)} = \frac{0.045}{0.140}.$$

$$\boxed{P(F|+) \approx 0.321.}$$

A positive diagnostic is not equivalent to a 90% failure probability because failure is rare a priori.

Part VIII. Statistics, Bayesian Updating, and Probability Distributions

Problem 71 — Sensitivity, specificity, and prevalence

A test has sensitivity 95.5% and specificity 99.5%. Find the probability of true infection after a positive test when prevalence is (a) 0.1% and (b) 1%.

Solution

The false-positive rate is 0.005. Bayes' theorem gives

$$P(D|+) = \frac{0.955P(D)}{0.955P(D) + 0.005[1 - P(D)]}.$$

For $P(D) = 0.001$, $P(D|+) \approx 0.1605$. For $P(D) = 0.01$, $P(D|+) \approx 0.6586$.

$$\boxed{16.1\% \text{ and } 65.9\%, \text{ respectively.}}$$

Problem 72 — Copolymer transition probability

In an observed sequence, an A unit is followed by B four times and by A six times. Estimate $P(B|A)$.

Solution

There are ten transitions beginning with A, four of which end in B. Therefore,

$$\boxed{\widehat{P}(B|A) = \frac{4}{10} = 0.40.}$$

Problem 73 — Expected chemical yield

A process gives yields 0%, 50%, and 100% with probabilities 0.10, 0.30, and 0.60. Find the expected yield.

Solution

$$E[Y] = 0(0.10) + 50(0.30) + 100(0.60) = \boxed{75\%}.$$

Problem 74 — Covariance and correlation of temperature and rate

For equally weighted observations $T = (300, 310, 320)$ and $r = (1, 2, 3)$, use population formulas to find covariance and correlation.

Solution

The centred vectors are $(-10, 0, 10)$ and $(-1, 0, 1)$. Therefore,

$$\mathrm{Cov}(T, r) = \frac{20}{3},$$

$$\mathrm{Var}(T) = \frac{200}{3}, \qquad \mathrm{Var}(r) = \frac{2}{3}.$$

Hence,

$$\boxed{\rho = 1.}$$

The data are perfectly linearly correlated.

Problem 75 — Variance of a combined measurement

Given $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = 3$, find $\mathrm{Var}(X + Y)$.

Solution

$$\mathrm{Var}(X + Y) = 4 + 9 + 2(3) = \boxed{19}.$$

Positive covariance increases the uncertainty of the sum.

Problem 76 — Gaussian measurement interval

A bond length follows $X \sim \mathcal{N}(10, 2^2)$. Approximate $P(8 \le X \le 12)$.

Solution

The interval is $\mu \pm \sigma$, corresponding to $-1 \le Z \le 1$. Therefore,

$$\boxed{P(8 \le X \le 12) \approx 0.6827.}$$

Problem 77 — Mahalanobis distance of a solvent descriptor

Let $x = (1, 2)^T$, $\mu = 0$, and $\Sigma = \mathrm{diag}(1, 4)$. Find the Mahalanobis and Euclidean distances.

Solution

$$d_M^2 = x^T \Sigma^{-1} x = 1 + \frac{4}{4} = 2,$$

so $d_M = \sqrt{2} \approx 1.414$. The Euclidean distance is $\sqrt{5} \approx 2.236$.

$$\boxed{d_M = \sqrt{2}, \qquad d_E = \sqrt{5}.}$$

Problem 78 — Conditional Gaussian prediction

Let (X, Y) have zero mean and covariance

$$\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}.$$

Find the conditional distribution of X given $Y = 1$.

Solution

$$\mu_{X|Y} = \Sigma_{XY} \Sigma_{YY}^{-1} Y = 2 \left(\frac{1}{3} \right) = \frac{2}{3}.$$

$$\Sigma_{X|Y} = 4 - 2 \left(\frac{1}{3} \right) 2 = \frac{8}{3}.$$

Therefore,

$$\boxed{X|Y=1 \sim \mathcal{N}\left(\frac{2}{3}, \frac{8}{3}\right).}$$

Problem 79 — Binomial catalyst-success count

Ten preparations succeed independently with probability 0.7. Find the probability of exactly seven successes.

Solution

$$P(X=7) = \binom{10}{7}(0.7)^7(0.3)^3 \approx \boxed{0.2668}.$$

Problem 80 — Beta update for conversion probability

A prior is $p \sim \text{Beta}(2,2)$. Eight successes and two failures are observed. Find the posterior and its mean.

Solution

The posterior is

$$p|\text{data} \sim \text{Beta}(2+8, 2+2) = \text{Beta}(10,4).$$

Its mean is

$$\boxed{E[p|\text{data}] = \frac{10}{14} = 0.714.}$$

Part IX. Chemical Machine-Learning Applications

Problem 81 — Linear regression for adsorption energy

A model is $E_{\text{ads}} = a + bx$. Training data are $(0,-1)$, $(1,-2)$, and $(2,-3)$. Determine a and b.

Solution

At $x = 0$, $a = -1$. The slope is

$$b = \frac{-2 - (-1)}{1 - 0} = -1.$$

Thus,

$$\boxed{E_{\text{ads}} = -1 - x.}$$

Problem 82 — Principal spectral direction

For

$$\Sigma = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix},$$

find the principal directions and the variance fraction captured by the first component.

Solution

The eigenvalues are 7 and 1. Corresponding normalized eigenvectors are $(1, 1)^T/\sqrt{2}$ and $(1, -1)^T/\sqrt{2}$. The first component captures

$$\frac{7}{7 + 1} = 0.875.$$

$$\boxed{87.5\% \text{ of the variance is captured by PC1.}}$$

Problem 83 — Feature scaling changes nearest-neighbor prediction

A query molecule is $Q = (1, 100)$. Library molecules are $A = (2, 105)$ and $B = (1.1, 130)$. Which is nearer before scaling? If descriptor scales are 1 and 50, which is nearer after scaling?

Solution

Raw distances are $\sqrt{26} \approx 5.10$ for A and about 30.0 for B, so A is nearer. Scaled distances are

$$d_s(A) = \sqrt{1^2 + 0.1^2} \approx 1.005,$$

$$d_s(B) = \sqrt{0.1^2 + 0.6^2} \approx 0.608.$$

$$\boxed{A \text{ is nearer before scaling; B is nearer after scaling.}}$$

Problem 84 — Equal Gaussian likelihood of two MD frames

A descriptor distribution has mean zero and covariance $\Sigma = \mathrm{diag}(1, 4)$. Compare the Gaussian likelihoods of $x_A = (1, 0)^T$ and $x_B = (0, 2)^T$.

Solution

Their Mahalanobis squared distances are

$$d_A^2 = 1, \qquad d_B^2 = \frac{2^2}{4} = 1.$$

Because the Gaussian normalization and exponent are identical,

$$\boxed{p(x_A) = p(x_B).}$$

Problem 85 — Bayesian phase classification

A material has phase priors $P(A) = 0.6$ and $P(B) = 0.4$. For an observed feature x, $p(x|A) = 0.2$ and $p(x|B) = 0.5$. Find $P(B|x)$.

Solution

$$P(B|x) = \frac{0.5(0.4)}{0.2(0.6) + 0.5(0.4)} = \frac{0.20}{0.32}.$$

$$\boxed{P(B|x) = 0.625.}$$

Problem 86 — Polynomial equation-of-state fitting

Fit $y = a + bx + cx^2$ to $(0, 1)$, $(1, 4)$, and $(2, 9)$.

Solution

At $x = 0$, $a = 1$. The remaining equations are $b + c = 3$ and $b + 2c = 4$, so $c = 1$ and $b = 2$.

$$\boxed{y = 1 + 2x + x^2.}$$

The model is nonlinear in x but linear in the unknown parameters.

Problem 87 — Dominant variance direction

For $\Sigma = \mathrm{diag}(9, 1)$, find PC1 and its explained-variance fraction.

Solution

The largest eigenvalue is 9 with eigenvector $(1, 0)^T$. The explained fraction is

$$\frac{9}{9 + 1} = 0.90.$$

$$\boxed{\text{PC1 is the first descriptor axis and explains 90\%.}}$$

Problem 88 — Gradient descent for a harmonic force constant

A force model is $\hat{F} = -kr$. Data are $(r, F) = (1, -3)$ and $(2, -6)$, with

$$L(k) = \frac{1}{2}[(-k + 3)^2 + (-2k + 6)^2].$$

Find $L'(k)$ and perform two updates from $k_0 = 0$ using $\gamma = 0.1$.

Solution

$$L'(k) = 5k - 15.$$

Thus $k_1 = 0 - 0.1(-15) = 1.5$. Then $L'(1.5) = -7.5$, so $k_2 = 1.5 + 0.75 = 2.25$.

$$\boxed{k_1 = 1.5, \qquad k_2 = 2.25.}$$

Problem 89 — Nonnegative spectral unmixing

Solve the nonnegative least-squares problem

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \approx \begin{bmatrix} 1 \\ 1.2 \end{bmatrix}, \qquad x, y \geq 0.$$

Solution

The exact unconstrained solution is $(x, y) = (1.2, -0.2)$, which is infeasible. Set the active variable $y = 0$ and minimize $(x-1)^2 + (x-1.2)^2$. The derivative gives $x = 1.1$.

$$\boxed{x = 1.1, \qquad y = 0.}$$

Problem 90 — Uncertainty propagation through a product

Independent measurements have means $(2, 3)$ and variances $(0.04, 0.09)$. For $y = x_1 x_2$, estimate $\mathrm{Var}(y)$ by first-order linearization.

Solution

At the mean, the Jacobian is $J = (3, 2)$. Therefore,

$$\mathrm{Var}(y) \approx J \begin{bmatrix} 0.04 & 0 \\ 0 & 0.09 \end{bmatrix} J^T = 9(0.04) + 4(0.09) = 0.72.$$

$$\boxed{\mathrm{Var}(y) \approx 0.72, \qquad \sigma_y \approx 0.849.}$$

Part X. Integrated and Advanced Chemical Applications

Problem 91 — Eigenmodes of reversible first-order kinetics

For

$$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B,$$

with $k_1 = 2$ and $k_2 = 1$, the population vector satisfies

$$\frac{d}{dt}\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} A \\ B \end{bmatrix}.$$

Find the eigenvalues and equilibrium ratio B/A.

Solution

The characteristic polynomial is $\lambda(\lambda + 3) = 0$, so the eigenvalues are 0 and -3. The zero-mode eigenvector satisfies $-2A + B = 0$, hence $B = 2A$.

$$\boxed{\lambda = 0, -3; \qquad B_{\text{eq}}/A_{\text{eq}} = 2.}$$

The nonzero mode relaxes with time constant 1/3.

Problem 92 — Mass weighting of a molecular Hessian

Given

$$H = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad M = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix},$$

construct $F = M^{-1/2} H M^{-1/2}$ and find its eigenvalues.

Solution

$$M^{-1/2} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix},$$

so

$$F = \begin{bmatrix} 2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda = \frac{2.5 \pm \sqrt{3.25}}{2},$$

or numerically

$$\boxed{\lambda_1 \approx 0.3486, \qquad \lambda_2 \approx 2.1514.}$$

Problem 93 — Detecting one dominant spectral component by SVD

For

$$X = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix},$$

find the rank and singular values.

Solution

The second row is one-half of the first, so the rank is 1. Since X is symmetric positive semidefinite, its singular values equal its eigenvalues. The trace is 5 and determinant is zero, giving

$$\boxed{\operatorname{rank}(X) = 1, \qquad \sigma_1 = 5, \qquad \sigma_2 = 0.}$$

A single latent spectral pattern can explain the matrix exactly.

Problem 94 — Removing a constant spectral baseline by projection

A spectrum is $y = (3, 4, 5)^T$. The baseline subspace is spanned by $\mathbf{1} = (1, 1, 1)^T$. Find the baseline and corrected spectrum.

Solution

The projection coefficient is

$$\frac{\mathbf{1}^T y}{\mathbf{1}^T \mathbf{1}} = \frac{12}{3} = 4.$$

Thus the baseline is $(4, 4, 4)^T$, and the corrected spectrum is

$$\boxed{y_{\text{corr}} = (-1, 0, 1)^T.}$$

The corrected spectrum is orthogonal to the constant baseline.

Problem 95 — Bayesian catalyst reliability after mixed outcomes

A catalyst success probability has prior $p \sim \mathrm{Beta}(5, 1)$. Three successes and two failures are observed. Find the posterior and posterior mean.

Solution

The posterior parameters are $\alpha' = 5 + 3 = 8$ and $\beta' = 1 + 2 = 3$.

$$\boxed{p|\text{data} \sim \mathrm{Beta}(8, 3).}$$

The posterior mean is

$$\boxed{E[p|\text{data}] = \frac{8}{11} \approx 0.727.}$$

Problem 96 — Combining two Gaussian sensors

Two independent sensors estimate the same concentration:

$$X_1 \sim \mathcal{N}(10, 4), \qquad X_2 \sim \mathcal{N}(12, 9).$$

Assuming a flat prior, find the fused Gaussian mean and variance.

Solution

Precisions add:

$$\frac{1}{\sigma^2} = \frac{1}{4} + \frac{1}{9} = \frac{13}{36},$$

so $\sigma^2 = 36/13 \approx 2.769$. The precision-weighted mean is

$$\mu = \sigma^2 \left(\frac{10}{4} + \frac{12}{9} \right) = \frac{138}{13} \approx 10.615.$$

$$\boxed{X_{\text{fused}} \sim \mathcal{N}(10.615, 2.769).}$$

Problem 97 — Linear transformation of a Gaussian descriptor vector

Let

$$x \sim \mathcal{N}\left(\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}1 & 0.5\\0.5 & 2\end{bmatrix}\right),$$

and define $y = (2, -1)x$. Find the distribution of y.

Solution

The mean is

$$E[y] = (2, -1)(1, 2)^T = 0.$$

The variance is

$$\mathrm{Var}(y) = (2, -1)\Sigma(2, -1)^T = 4.$$

Therefore,

$$\boxed{y \sim \mathcal{N}(0, 4).}$$

Problem 98 — Projection onto a feed constraint

An unconstrained preferred feed composition is $(4, 2)$. The plant requires $x+y = 5$. Find the feasible composition closest in Euclidean distance.

Solution

The preferred composition has total 6, so it must be shifted by one unit along the normal direction $(1, 1)$. Divide the correction equally:

$$\Delta x = \Delta y = -\frac{1}{2}.$$

Hence,

$$\boxed{(x, y) = (3.5, 1.5).}$$

This is the orthogonal projection onto the constraint line.

Problem 99 — A rank-3 polarizability derivative tensor

Let

$$\alpha(E_1, E_2) = \begin{bmatrix} 1 + 2E_1 & E_2 \\ E_2 & 3 - E_1 \end{bmatrix}.$$

Find $\partial\alpha/\partial E_1$ and $\partial\alpha/\partial E_2$.

Solution

Differentiate elementwise:

$$\boxed{\frac{\partial\alpha}{\partial E_1} = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \frac{\partial\alpha}{\partial E_2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.}$$

Together they form the third-rank tensor $\partial\alpha_{ij}/\partial E_k$.

Problem 100 — Integrated PCA and regression for molecular data

Three molecules have

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}, \qquad y = (-1, -2, -3)^T.$$

Using population normalization $1/N$, centre X, compute its covariance, find PC1, and regress centred y on the PC1 scores.

Solution

The mean descriptor is $(2, 2)$, so

$$X_c = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}.$$

The covariance is

$$\Sigma = \frac{1}{3} X_c^T X_c = \begin{bmatrix} 2/3 & 2/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

Its eigenvalues are $4/3$ and 0, with

$$u_1 = \frac{1}{\sqrt{2}}(1,1)^T.$$

The PC1 scores are $t = (-\sqrt{2}, 0, \sqrt{2})^T$. Since $\bar{y} = -2$, the centred target is $(1, 0, -1)^T$. The regression slope through the centred origin is

$$b = \frac{t^T y_c}{t^T t} = -\frac{1}{\sqrt{2}}.$$

Therefore,

$$\boxed{\hat{y} = -2 - \frac{1}{\sqrt{2}}t.}$$

This problem combines centring, covariance, eigenvectors, dimensionality reduction, projection, and regression.

Compact NumPy Verification Template

```python
import numpy as np

# Linear system
A = np.array([[1.0, 0.5],
              [0.2, 1.0]])
b = np.array([1.2, 0.8])
x = np.linalg.solve(A, b)

# Least squares
coef, residuals, rank, singular_values = np.linalg.lstsq(A, b, rcond=None)

# Eigenproblem
H = np.array([[2.0, 1.0],
              [1.0, 2.0]])
evals, evecs = np.linalg.eigh(H)

# Singular-value decomposition
U, s, VT = np.linalg.svd(A, full_matrices=False)

# Covariance and PCA
X = np.array([[1.0, 1.0],
              [2.0, 2.0],
              [3.0, 3.0]])
```

```
Xc = X - X.mean(axis=0)
Sigma = Xc.T @ Xc / len(X)
evals, evecs = np.linalg.eigh(Sigma)
```

End of problem set